

CONFLUENCE × OnCo: An Evidence-Gated Dynamical Framework for Integrating Oncology Knowledge Graphs with Adaptive Cancer-State Models

Thesis #1 - computational research thesis

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This manuscript expands ONCO_CONFLUENCE_THESIS_FINDINGS.md, docs/ONCO_CONFLUENCE_ontology_ and docs/ONCO_ADAPTER.md in Project Confluence. It does not add wet-lab measurements, patient-level results, or invented identifiers. It is not the 2022 B.Sc. Carica papaya AgNP antidiabetic project; that wet-lab thesis is catalogued separately as Thesis 0.

Title page

CONFLUENCE × ONCO: AN EVIDENCE-GATED DYNAMICAL FRAMEWORK
FOR INTEGRATING ONCOLOGY KNOWLEDGE GRAPHS WITH ADAPTIVE
CANCER-STATE MODELS

BY

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(IN-SILICO / ARCHITECTURAL STUDY)

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Declaration

I, Kelechi Emeka Ogbonna, hereby declare that this computational research thesis titled “CONFLUENCE × OnCo: An Evidence-Gated Dynamical Framework for Integrating Oncology Knowledge Graphs with Adaptive Cancer-State Models” was carried out by me. The findings reported here are in-silico and architectural. They are not wet-lab measurements, not patient outcomes, and not a claim of cure, dose, or clinical decision support. No identifier (DOI, ORCID, journal acceptance) has been invented.

Kelechi Emeka Ogbonna Date

Abstract

Cancer research now produces knowledge graphs, multi-omic assays and dynamical simulators in parallel [10,53,61]. The failure mode is collapsing those layers. This study asks whether a provenance-controlled pipeline can keep the layers apart while still allowing testable predictions. Scientific success is a staged chain – traceability, mathematical validity, identifiability, out-of-sample prediction, experimental falsification – not disease eradication [43-52].

CONFLUENCE v2 is a frozen 15-dimensional computational cancer-state model with an optional connectome-style controller [11,58,60]. OnCo is a public, cited oncology knowledge graph [10,61]. They are complementary only if knowledge is not written into parameters [57,58]. Pull request #9 shipped a read-only OnCo adapter and nine refusal tests. It is architectural evidence for the gates, not therapeutic efficacy [11,56].

GLOBOCAN 2024 estimates, published 2026, place the contemporary burden at about 20.6 million diagnoses and 9.8 million deaths [1,2,64]. That statistic motivates computational oncology; it is not a CONFLUENCE parameter [1,59]. This document records the gates, the P0 adapter, the LDHA / p_{lactate} refusal, and the honesty rule that OnCo confidence is not P(H) [10,11,36,63].

The key objectives of this research study are to:

- i. Forbid skip-level promotion from OnCo knowledge to ordinary-differential-equation (ODE) parameters
- ii. Classify legacy artefacts, including `validation/gene_to_parameter_map.json`, as assumed / unidentified rather than as identified Θ
- iii. Ship and audit a read-only P0 adapter without editing CancerODE
- iv. Keep the OnCo Ideas shelf separate from any CONFLUENCE hypothesis library
- v. Define success as gated, falsifiable computational work, not a universal cure

The methodology in this study is computational. Materials are public software artefacts, cited knowledge-graph records, and numbered Vancouver references. No patient data were used. No wet-lab assay was performed.

Findings from this study revealed that the honesty ladder holds as an object model; that `refuse_knowledge_as_parameter(OncoRef("ldha"), "p_lactate")` returns provenance forbidden; that the legacy gene-to-parameter map remains assumed / unidentified; and that pull request #9 does not edit CancerODE. No new parameter was identified. No patient outcome is claimed.

This thesis is research only. It is not a medical device, not clinical decision support, and not a wet-lab antidiabetic study.

Keywords

computational oncology; knowledge graphs; dynamical systems; evidence gates; identifiability; tumour heterogeneity; tumour microenvironment; OnCo; CONFLUENCE; research-only; not a medical device

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Results in Chapter Four are computational audits, gate outcomes, and refusal cases. They are not fabricated patient outcomes.

CHAPTER ONE

1.0 INTRODUCTION

1.1 Background to the study

Oncology knowledge graphs and dynamical models

Oncology knowledge graphs name genes, diseases, ideas and citations [10,53,61]. Dynamical simulators ask how a state moves under control [8,39,43,44]. Mixing the two without provenance treats a web page as an identified parameter [45-52,57]. The problem stated on the public thesis page, and restated here, is: how can heterogeneous oncology knowledge and molecular observations be incorporated into a cancer dynamical model without collapsing evidence, mechanism, parameterisation and prediction into unsupported assumptions [57,59]?

GLOBOCAN 2024 estimates, published 8 July 2026, report about 20.6 million diagnoses and 9.8 million deaths across 34 cancer types in 186 countries [1,2,64]. Female breast cancer accounted for about 2.43 million new cases in that estimate series [1,2]. Those figures continue a modelled GLOBOCAN series previously reported for 2022 and 2020 [54,55]. WHO states that many cancers can be cured if detected early and treated effectively, while access remains uneven [3]. That sentence is a policy statement about staged, treatable disease – not a claim that CONFLUENCE or OnCo cures patients [3,10,11,56]. The WHO global status report on cancer 2026 is cited as health-system context only [4]. Nigeria GLOBOCAN 2022 estimates (127,763 new cases; 79,542 deaths; breast 32,278) are setting context only; they are not 2026 incidence [5].

Cancer as an adaptive, microenvironment-coupled process

Cancer is organised as a set of acquired capabilities – the hallmarks – rather than as a single targetable defect [13-15]. Tumours evolve clonally and adapt under therapy [7,16,17]. Intra-tumour heterogeneity, branched evolution and therapy resistance are documented empirical features of that process [18-21]. Multi-omics increase resolution without automatically producing a causal model [20,43,49]. The tumour microenvironment recruits stromal and immune cells that regulate progression, exclusion and metastasis [22-27]. Mathematical oncology supplies in-silico laboratories that couple those layers as dynamical systems [8,39]. Adaptive therapy treats treatment as a process under selection rather than as a fixed maximum-tolerated pulse [9,40,41]. The gap this thesis addresses is connecting knowledge to dynamical hypotheses without dropping provenance [10,11,57].

TNBC as a computational test case, not a treated cohort

Triple-negative breast cancer (TNBC) is used as a test case because NCI describes it as roughly 15% of breast cancers, typically faster-growing and more recurrent, and heterogeneous [6]. Independent reviews describe the same clinical heterogeneity and recurrence pattern [28-31]. That description is not a CONFLUENCE parameter [6,56,63]. No patient with TNBC was enrolled, treated, or simulated as a named person in this study.

1.2 STATEMENT OF RESEARCH PROBLEM

How can heterogeneous oncology knowledge-graph records and molecular observations be incorporated into a frozen cancer dynamical model without collapsing knowledge, evidence, mechanism, parameterisation, and prediction into unsupported assumptions [10,45-53,57,61]?

That is the research problem. It is a computational and medical-methods problem. It is not a claim to treat, dose, or cure patients [3,56].

Cancer research now produces knowledge graphs, multi-omic assays, and dynamical simulators in parallel [10,20,43,44,53,61]. OnCo names genes, diseases, ideas, and citations [10,61]. CONFLUENCE v2 asks how a frozen 15-dimensional state moves under an in-silico controller [11,58,60]. Mixing the two without provenance treats a web page, an OnCo confidence score, or a legacy numeric map as an identified parameter Θ [45-52,57,63]. The conversion ladder - Knowledge -> Evidence -> Hypothesis -> Mechanism -> Parameter -> Prediction -> Experiment -> new Evidence - has a failure point at every arrow [57]. A literature record may motivate a hypothesis; it must not become Θ [45,47,51,57].

The problem is therefore to keep those layers as typed objects, to refuse skip-level promotion (including LDHA as `p_lactate`), and to define scientific success as a staged, falsifiable chain rather than as disease eradication [11,12,36,43-52,56,58].

1.3 JUSTIFICATION OF STUDY

Existing integration habits fail in documented ways that this study is built to catch.

Overclaiming. GLOBOCAN 2024 estimates (about 20.6 million diagnoses and 9.8 million deaths) and WHO language that many cancers can be cured if found early and treated well are health-system context [1-4,54,55,64]. They are not CONFLUENCE parameters and not a product claim [56,59]. Adaptive-therapy and mathematical-oncology papers supply in-silico laboratories and control ideas; they are not CONFLUENCE efficacy [8,9,39-41]. Biomedical knowledge graphs integrate assertions; they remain literature objects until an identification step is performed [53].

Parameter smuggling. The P0 findings record the failure mode: treat OnCo's LDHA lactate-metabolism listing as the v2 symbol `p_lactate`, or treat `validation/gene_to_parameter_map.json` (LDHA -> `pyruvate_to_lactate` (+0.10 / +0.30)) as identified Θ [10,11,36,58,63]. Alias trap: legacy `pyruvate_to_lactate` is not v2 `p_lactate` (default 0.22) [58,63]. Warburg and lactate-immune papers explain why the gene is scientifically interesting; they do not identify the coefficient

[32-38,42]. OnCo confidence.probability is not $P(H)$. Idea maturity is not evidence level [10,57,59].

Missing gates. Structural identifiability asks whether a unique parameter vector is consistent with noise-free input-output data [45,46,48,50]. Practical identifiability asks whether finite, noisy data actually constrain those parameters [47,52]. Nonlinear biological systems are often only partially observable; unidentified symbols must remain labelled as such [51]. Without Gate 4, a silent write from a knowledge record into CancerODE produces a confident but unidentifiable model [43-52,57]. Hallmarks, clonal evolution, intra-tumour heterogeneity, and the tumour microenvironment further explain why a named gene or a TNBC subtype label is not a sufficient statistic for Θ [6,7,13-21,22-31].

What this study therefore does. Pull request #9 shipped a read-only OnCo adapter and nine refusal tests without editing CancerODE [11,58]. This manuscript records those gates as architectural evidence, not as therapeutic efficacy [11,56].

1.4 AIM AND OBJECTIVES OF THE STUDY

The aim of this research is to specify and audit an evidence-gated computational framework that can bind a public oncology knowledge graph (OnCo) to a frozen cancer dynamical model (CONFLUENCE v2) without treating a cited page, a confidence score, or a legacy numeric map as an identified parameter.

The objectives of this research study are to:

- i. Forbid skip-level promotion from OnCo knowledge to ODE parameters [10,11,57,58]
- ii. Classify legacy artefacts, including validation/gene_to_parameter_map.json, as assumed / unidentified rather than as identified Θ [47,51,52,63]
- iii. Ship a read-only P0 adapter without editing CancerODE [11,58]
- iv. Keep the OnCo Ideas shelf separate from any CONFLUENCE hypothesis library [10,61]
- v. Define success as gated, falsifiable computational work – control and disease-specific clearance – not a universal cure [3,12,56]

1.5 SIGNIFICANCE OF THE STUDY

Scientific significance for researchers. The work treats cancer as a dynamic, adaptive, microenvironment-coupled process rather than as a bag of independent named targets [7,13-17,22-27]. It gives laboratories a typed conversion ladder and a success criterion – traceability, mathematical validity, identifiability, out-of-sample prediction, experimental falsification – that can fail honestly [12,43-52,57]. Failed or refused claims (LDHA as p_lactate) stay visible [11,58,63].

Methodological significance. Evidence gates and the identifiability literature become refusal rules, not a new structural-identifiability proof for confluence_v2_15d [45-52]. Layer objects (OncoRef, Evidence, Mechanism, Parameter, Prediction) may point at the next layer; they may not collapse into it [57]. Public claims require a ledger id or a numbered Vancouver reference; DOIs are not invented [59].

What this significance is not. It is not clinical decision support, not a medical device, not a dosing table, and not a cure [3,56]. GLOBOCAN burden figures motivate computational oncology; they do not licence a CONFLUENCE product [1,2,56,64]. Translation waits for Gates 6-8, which this repository does not claim to have passed [3,12,56]. The significance of the study is methodological honesty for researchers, not a path to a clinic. It is also not a wet-lab claim about plant extracts or silver nanoparticles.

1.6 SCOPE OF THE STUDY

This study is limited to computational architecture and software-gate audit on Project Confluence artefacts. The scope covers: the P0 OnCo adapter; the five-layer object model; the conversion ladder; refusal of LDHA as p_lactate; and classification of the legacy gene-to-parameter map as unidentified.

The scope does not include: patient recruitment; wet-lab assay; ODE refit; automatic fitting; 3-state ROS import; 6-state expansion; clinical trial; RECIST adjudication; or a dosing table [12,56]. GLOBOCAN and WHO figures are setting context only [1-5,64]. Nigeria 2022 estimates are not 2026 incidence [5]. This document has no DOI.

CHAPTER TWO

2.0 LITERATURE REVIEW

2.1 Oncology knowledge graphs and dynamical simulators

OnCo answers what is named, linked, dated and cited [10,61]. Biomedical knowledge graphs more generally integrate heterogeneous assertions; they remain literature objects until an identification step is performed [53]. CONFLUENCE v2 answers how a frozen 15-D state moves under an in-silico controller [11,58,60]. The working hypothesis of this thesis is that those artefacts are complementary provided they are not collapsed into each other [57].

OnCo Ideas carry hypothesis, rationale, test and maturity fields [10,61]. Bulk ingest into CONFLUENCE is forbidden [57,58]. Idea maturity is not evidence level. OnCo confidence.probability is not $P(H)$ [57,59]. OnCo integration does not justify changing the dynamics [11,58].

2.2 Hallmarks, heterogeneity and the tumour microenvironment

Hanahan and Weinberg organised neoplastic growth as a small set of acquired capabilities, later expanded and re-dimensioned [13-15]. That organising frame is why a knowledge graph of named targets is not a closed dynamical explanation: the same hallmark can be implemented by different molecular routes [14,15,20]. Clonal evolution and ecological competition supply the time axis [16,17]. Intra-tumour heterogeneity is both a looking-glass for those processes and a documented source of therapeutic failure [7,18-21].

The tumour microenvironment is not a passive backdrop. Recruited stromal and immune populations regulate progression and metastasis [22-24]. T-cell exclusion and oncogenic-pathway-driven immune evasion are review-level mechanisms for why a metabolic or antigenic fact does not become a control law [25,26]. Exhaustion is a distinct T-cell state, not a synonym for “the immune variable is low” [27]. None of those reviews is treated here as an identified CONFLUENCE coupling.

2.3 Warburg metabolism and lactate as research context

Aerobic glycolysis is a classical observation in cancer cells [32]. Later work reframed the Warburg effect as a proliferation-linked metabolic programme rather than a simple mitochondrial defect [33-35]. Lactate is not only a waste product: tumour-derived

lactic acid can polarise macrophages and blunt T- and NK-cell surveillance [36-38]. A microenvironmental model of carcinogenesis treats acidification and selection as coupled processes [42].

OnCo lists LDHA among lactate-metabolism targets [10]. The CONFLUENCE v2 symbol `p_lactate` is a frozen-model parameter. The P0 adapter refuses to write the OnCo record into that symbol [11,36,58,63]. The literature explains why the gene is scientifically interesting. It does not identify Θ .

2.4 Mathematical oncology and adaptive control

Ordinary differential equations, stochastic models, agent-based approaches and control-theoretic methods can represent tumour growth, treatment response and resistance as in-silico laboratories [8,39]. Adaptive therapy, introduced as an evolutionary alternative to continuous maximum-tolerated dosing, treats sensitive clones as competitive suppressors of resistant clones [9,40]. A later clinical-research programme in metastatic castrate-resistant prostate cancer is cited only as context that the control idea has a literature, not as a CONFLUENCE protocol or a patient result [41]. CONFLUENCE controllers that implement a Gatenby-style treat-and-halt heuristic remain in-silico research arms [9,12,56].

2.5 Identifiability of dynamical biological models

Systems biology treats a model as a formal object whose parameters may or may not be recoverable from observations [43,44]. Structural identifiability asks whether a unique parameter vector is consistent with noise-free input-output data [45,46,48,50]. Practical identifiability asks whether finite, noisy data actually constrain those parameters [47,52]. Nonlinear biological systems are often only partially observable; unidentified symbols must remain labelled as such [51]. Reverse engineering and identification remain open strategies with stated limits [49]. This literature is used in Chapter Three as Gate 4, not as a new identifiability computation for `confluence_v2_15d`.

2.6 Frozen CONFLUENCE lineages and OnCo Ideas

Frozen lineages in this repository are:

Table 2-1. Frozen CONFLUENCE lineages

ID	Role	Status
<code>tnbc_mod_3s</code>	TNBC-Metabolic-Strain-MOD notebooks	frozen; ROS audit pending [62]
<code>confluence_report_6s</code>	report architecture $X = [T, I, S, L, R, H]$	paper only [60]
<code>confluence_v2_15d</code>	live 15-D CancerODE	adapter sits around it [11,58]
<code>confluence_v1_calibrator</code>	<code>gene_to_parameter_map.json</code>	executable, not identified [63]

Table 2-2. Layer objects (ontology spec v0.3)

The ingestion spec keeps five objects [57]. A record may point at the next layer. It may not collapse into it.

Layer	Object	Allowed question	Forbidden leap
Knowledge	OncoRef	What does OnCo name and link?	therefore k = ...
Evidence	EvidenceObject	What was measured, where, in what system?	therefore this term belongs in F
Causal mechanism	MechanismObject	do(U) changes which state, in which context?	therefore identifiable from CCLE
Parameter	ParameterObject	Which symbol in which frozen model?	silent write into rhs_cancer
Prediction	PredictionObject	Frozen-model output under U, with uncertainty	clinical advice

Figure 2-1. Conversion ladder

OnCo knowledge

```
--cite--> Evidence      (source URI + rung required)
--interpret--> Hypothesis (falsifier required)
--propose--> Mechanism  (context + sign + do-operator)
--identify--> Parameter (model_id + symbol + identifiability ≠ unidentified)
--simulate--> Prediction
--test--> Experiment
--write--> Evidence
```

Every arrow is a failure point. A page may motivate a hypothesis; it must not become Θ [45,47,51,57]. No arrow may skip a box. Wired RHS terms remain Parameters with provenance assumed until identified [52,63].

legacy gene->parameter map != identified parameter mapping. Alias trap: legacy pyruvate_to_lactate is not v2 p_lactate (default 0.22) [58,63].

CHAPTER THREE

3.0 MATERIALS AND METHODS

3.1 Study design and honesty statement

This is a computational methods and architecture study. There is no human-subjects protocol, no animal protocol, no chart review, and no wet-lab bench work. Inputs are public software artefacts, cited OnCo records, and numbered Vancouver references. Outputs are gate audits, adapter refusal cases, and this manuscript. The analysis unit is the **software object**, not the patient.

Materials:

- Project Confluence software and documentation (confluence/onco/, docs/ONCO_ADAPTER.md, ontology spec v0.3) [11,57,58,60]
- OnCo public knowledge graph (data CC BY-NC 4.0) [10,61]
- Legacy map validation/gene_to_parameter_map.json [63]
- Validation protocol validation/validation_protocol.md [12]
- Honesty gates as stated in DISCLAIMER.md [56]

Results in Chapter Four are therefore in-silico / architectural. They must not be read as patient outcomes or as wet-lab antidiabetic findings.

3.2 Honesty gates

The public protocol is sequential and falsifiable [12,56,59]. It is not a treatment path.

Table 3-1. Honesty gates

Gate	Requirement
0	Research-only scope. Not a medical device. Not clinical decision support. [56]
1	Provenance: source, URI, date, population. [59]
2	Evidence class named. [57]
3	Mechanism: intervention, state, sign, context, falsifier. [57]

Gate	Requirement
4	Parameter: model id, symbol, units, estimator, uncertainty, identifiability. Unidentified != established. [45-52]
5	Prediction labelled as computational. [56]
6	Held-out / external / baseline validation. [12]
7	An experiment that could show the prediction is wrong. [12]
8	Translation only after biological and clinical validation this repository does not claim. [3,56]

Scientific success is: traceability -> mathematical validity -> identifiability -> out-of-sample prediction -> experimental falsification [43,44,51,52].

3.3 Identifiability as a methods claim, not a result

CONFLUENCE Gate 4 is the identifiability literature applied as a refusal rule: OnCo confidence and the legacy numeric map are not estimators [10,47,63]. This manuscript does not report a new identifiability computation.

3.4 Adapter P0

confluence/onco/ is a read-only client: cache envelope, bindings, schemas [11,58]. Commands documented in docs/ONCO_ADAPTER.md include fixture meta, bind --id ldha, and wired [58]. bind() returns slot annotations. It never writes p_lactate or pyruvate_to_lactate [11,63]. refuse_knowledge_as_parameter returns provenance forbidden [11,57]. Tests live in tests/test_onco_adapter.py [11,60].

OnCo data is CC BY-NC 4.0 [10,61]. Adapter code in this repository is MIT [60]. Cached payloads are not vendored; live cache stays under data/onco/cache/ (gitignored) [58]. Attribution on every export: Data from OnCo (onco.cc), CC BY-NC 4.0; commercial use needs a licence [10,59].

P0 ships: read-only client + cache envelope + bindings + nine gates [11,58]. P0 does not ship: RHS edits, controller edits, 3-state ROS import, 6-state expansion, automatic fitting, reverse writes to OnCo [11,58,62].

3.5 Refusal rules

1. Do not vendor the OnCo corpus [10,61].
2. OnCo is not a parameter source. refuse_knowledge_as_parameter(OncoRef("ldha"), "p_lactate") returns provenance forbidden [11,36,58].

3. OnCo is not a controller prior [9,40,58].
4. Wired RHS terms remain Parameters with provenance assumed until identified [47,52,63].
5. No reverse writes to OnCo [10,61].
6. No ODE right-hand-side edits in the P0 / thesis-#1 scope [11,58].
7. Public claims need a ledger id or a numbered reference (docs/CITATION_POLICY.md) [59].
8. Do not fabricate DOIs [59].

This manuscript uses the Vancouver entries generated from docs/manuscript/thesis_01_bibliography. Every doi: field was checked against Crossref or PubMed on 20 September 2026. Agency and GitHub items are URL citations without a DOI [2-7,10-12,56-64].

3.6 What was not done

No patient data. No ODE refit. No automatic fitting. No 3-state ROS import. No 6-state expansion. No clinical trial, RECIST adjudication, or dosing table [12,56]. No phytochemical assay, nanoparticle synthesis, or alpha-amylase inhibition study was performed. Those belong to a separate wet-lab B.Sc. project and are not mixed into this computational thesis.

CHAPTER FOUR

4.0 RESULTS

These are specification and software-architecture findings. They are not experimental oncology results and not patient outcomes [56]. Tables and figures below are computational audits, gate outcomes, and refusal cases.

4.1 Honesty ladder as an object model

Knowledge, Evidence, Causal mechanism, Parameter and Prediction are separate types [57]. A record may point at the next layer. It may not collapse into it.

4.2 LDHA is not p_lactate (refuse-as-parameter)

`refuse_knowledge_as_parameter(OncoRef("ldha"), "p_lactate")` is forbidden [11,58]. The auditor demo claim - that LDHA expression can be entered as `p_lactate` because OnCo lists LDHA as a lactate-metabolism target - is the claim the gate is built to reject [10,32,36,63].

4.3 Legacy map is assumed / unidentified

`validation/gene_to_parameter_map.json` maps LDHA $\rightarrow$ pyruvate_to_lactate (+0.10 / +0.30) [63]. That row is a legacy numeric hint, not an identified v2 parameter [47,51,52].

4.4 No ODE writes

PR #9 and the P0 adapter do not edit `CancerODE.rhs_cancer` [11,58].

4.5 OnCo is a cited graph, not Θ

OnCo is a cited graph, not Θ [10,61]. Counts move with `buildDate`.

4.6 Three-arm programme remains a protocol

`validation/validation_protocol.md` states H0 versus H1. Thresholds such as $r > 0.5$ are criteria, not findings [12].

4.7 Public ledger

Public ledger keeps year and population. Claims EVID-001-EVID-010 on the thesis page cite [1-12] without promoting burden statistics into model parameters [1,5,59,64].

No wet-lab measurement is reported here. No new parameter was identified [45,52,56].

CHAPTER FIVE

5.0 DISCUSSION, CONCLUSION AND RECOMMENDATION

5.1 Discussion

Keeping OnCo and CONFLUENCE apart is the result [10,11,57]. Integration is valuable only as a gated pipeline: cite -> interpret -> propose -> identify -> simulate -> test [49,57]. Collapsing any two boxes produces a confident but unidentifiable model [45,47,51].

The hallmarks, clonal-evolution and microenvironment literatures explain why a named gene, a lactate observation, or a TNBC subtype label is not a sufficient statistic for Θ [13-15,16-21,22-27,28-31]. Warburg and lactate-immune papers explain why LDHA appears in OnCo and why lactate is biologically non-trivial; they still do not license a silent write into p_{lactate} [32-38,42]. Mathematical oncology and adaptive-therapy papers explain why a controller layer can exist around a frozen model; they are not CONFLUENCE efficacy [8,9,39-41].

WHO language that many cancers can be cured if found early and treated well is retained as a health-system statement [3,4]. It is not transferred to CONFLUENCE as a product claim [56]. Prior GLOBOCAN editions (2020, 2022) are cited so that the 2024/2026 series is not treated as a one-off number [1,54,55,64].

Disease profiles and the thinking lab are scaffolds for questions. They are not protocols and are not the Scholar landing page [56,59]. The Scholar article URL is /thesis with the same-directory PDF /thesis.pdf.

Do not expand the state vector merely because OnCo lists more cell types [10,23,61]. Cache has no TTL yet [58]. Those are engineering limits, not clinical limits [56].

Limitations of this study, stated so they cannot be skipped:

- No new wet-lab measurement [56].
- PR #9 does not identify p_{lactate} or any other Θ [11,52].
- The 3-state ROS notebook remains unaudited [62].
- The three-arm protocol can fail; that would still be a result [12].

- OnCo counts move with buildDate [10].
- The optional Grok auditor may miss sources; “insufficient” is preferred to invented citations [59].
- Nigeria 2022 estimates are setting context only [5].
- This document has no DOI.
- Identifiability citations define Gate 4; they are not a new structural-identifiability proof for confluence_v2_15d [45-52].

5.2 Conclusion

The aim of this research was to bind OnCo knowledge to a frozen CONFLUENCE dynamical model without collapsing knowledge into parameters. The study concludes that:

- i. The five-layer object model and conversion ladder can be stated as typed, refuseable contracts [57].
- ii. The P0 adapter refuses LDHA as p_lactate and does not edit CancerODE [11,58].
- iii. The legacy gene-to-parameter map remains assumed / unidentified [63].
- iv. Scientific success for this thesis is architectural honesty, not disease eradication [3,12,43-52,56].

This work is computational research. It is not a medical device, not CDS, not a dose, and not a cure. It is not the 2022 wet-lab Carica papaya AgNP antidiabetic project.

5.3 Recommendation

The following recommendations follow from the gates, not from a clinic:

- i. Iterate the adapter without RHS edits [11,58].
- ii. Admit-list a hypothesis library separate from OnCo Ideas [10,61].
- iii. Identify p_lactate only after gates 1-4 stay green and an estimator with uncertainty exists [47,52,58].
- iv. Complete external / held-out validation (Gate 6) before any experimental falsifier (Gate 7) [12].
- v. Optional later: deposit this PDF on Zenodo or a preprint server and only then add citation_doi to the HTML landing page.

- vi. Translation (Gate 8) should remain out of scope until biological and clinical validation that this repository does not claim [3,56].
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Journal items use Vancouver form. DOI fields appear only for Crossref- or PubMed-verified journal records. WHO, IARC, NCI and GitHub items are complete Internet citations without a doi: field. No DOI is invented.

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